

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5	Assessment:	Assessment Tool 1
Weightage:	40% of CIA	Total Marks:	100
CO5 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	CSE-AI	Date:	26/08/2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G.VEDAVIKAS GOWD

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

CO5-AT1: Pseudocode Design

CART Classification Algorithm using Gini Impurity

1. Aim

To design and implement the CART (Classification and Regression Tree) classification algorithm to select the best binary split using Gini Impurity and construct a classification tree.

2. Algorithm

1. Start with the complete training dataset.
2. Calculate the Gini impurity for possible binary splits.
3. For each feature, test possible split values.
4. Divide the data into left and right groups.
5. Calculate the weighted Gini impurity of each split.
6. Select the split with the minimum Gini impurity.
7. Repeat the process recursively for child nodes.
8. Stop when the node is pure, maximum depth is reached, or no useful split exists.
9. Assign the majority class to each leaf node.
10. Use the constructed tree to classify new data.

3. Pseudocode

BEGIN

FUNCTION Gini(groups, classes)

 total_samples $\leftarrow$ total number of samples in all groups

 gini $\leftarrow$ 0

 FOR each group IN groups

 size $\leftarrow$ number of samples in group

 IF size = 0

 CONTINUE

 END IF

 score $\leftarrow$ 0

 FOR each class IN classes

 proportion $\leftarrow$ number of samples of class / size

 score $\leftarrow$ score + proportion $\times$ proportion

 END FOR

 gini $\leftarrow$ gini + (1 - score) $\times$ (size / total_samples)

 END FOR

RETURN gini

END FUNCTION

FUNCTION BestSplit(dataset)

 best_gini $\leftarrow$ infinity

 best_feature $\leftarrow$ NULL

 best_value $\leftarrow$ NULL

 FOR each feature IN dataset

 FOR each possible value IN feature

 Split dataset into Left and Right groups

 gini $\leftarrow$ Gini(Left, Right)

 IF gini < best_gini

 best_gini $\leftarrow$ gini

 best_feature $\leftarrow$ feature

 best_value $\leftarrow$ value

 END IF

 END FOR

 END FOR

 RETURN best_feature, best_value

END FUNCTION

FUNCTION BuildTree(dataset)

 IF all samples belong to same class

 RETURN Leaf Node

 END IF

 feature, value $\leftarrow$ BestSplit(dataset)

 Split dataset using feature and value

 left_tree $\leftarrow$ BuildTree(left group)

 right_tree $\leftarrow$ BuildTree(right group)

 RETURN Decision Node(feature, value, left_tree, right_tree)

END FUNCTION

END

4. Python Code

```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, export_text

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create CART classification tree using Gini impurity
model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = model.score(X_test, y_test)

print("CART Classification Tree")
print("-----")
print("Accuracy:", accuracy)

# Display tree rules
tree_rules = export_text(
    model, feature_names=list(iris.feature_names)
)
print("\nDecision Tree Rules:")
print(tree_rules)

# Predict one sample
```

```
sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(sample)

print("Predicted Class:", iris.target_names[prediction[0]])
```

5. Input

Dataset: Iris Dataset

Sample input:

Sepal Length = 5.1

Sepal Width = 3.5

Petal Length = 1.4

Petal Width = 0.2

The input features are used to determine the best binary split and classify the sample.

6. Output

CART Classification Tree

Accuracy: 1.0

Predicted Class: setosa

Example decision rule:

petal length ≤ 2.45

→ class: setosa

petal length > 2.45

→ further binary splits

7. Result

The CART classification algorithm was successfully implemented using Gini impurity. The algorithm selected the best binary splits, constructed a decision tree, and successfully classified the input data into the appropriate classes.